

List of experiments for DC Practical Exam

1. Write a simple **client-server application using RPC** where the client sends two integers, and the server returns their sum. Compare performance and behaviour of a local function call vs. an RPC call with timing outputs.
2. Write a **RMI application** that allows a client to perform basic arithmetic operations (add, subtract, multiply, divide) on the server.
3. Write **Berkeley's Clock Synchronization Algorithm**. Provide a detailed example to illustrate how the algorithm synchronizes the clocks of different nodes in the system.
4. Write the **Bully Algorithm** for electing a new coordinator in a distributed system. Provide a detailed example to demonstrate how the algorithm works and elects a new coordinator.
5. Write the **Ring Algorithm** for electing a new coordinator in a distributed system. Provide a detailed example to demonstrate how the algorithm works and elects a new coordinator.
6. Write **Ricart Agrawala's Algorithm** for achieving mutual exclusion in a distributed system. Support your explanation with an example.
7. Write **Raymond's Tree-Based Algorithm** for achieving mutual exclusion in a distributed system. Support your explanation with an example.
8. Write **Chandy-Misra-Haas algorithm** for distributed deadlock detection. Provide a detailed example to demonstrate how the algorithm detects a deadlock in a distributed system.
9. Write an algorithm for dynamic load balancing using the **Least Connections method**. Provide a detailed example to illustrate how the algorithm works in practice.
10. Develop an algorithm for static load balancing using the **Weighted Round Robin method**. Include a step-by-step example to demonstrate how tasks are distributed among servers based on their weights.